

Pointing the Unknown Words

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Abstract

The problem of rare and unknown words is an important issue that can potentially effect the performance of many NLP systems, including both the traditional count-based and the deep learning models. We propose a novel way to deal with the rare and unseen words for the neural network models using attention. Our model uses two softmax layers in order to predict the next word in conditional language models: one predicts the location of a word in the source sentence, and the other predicts a word in the shortlist vocabulary. At each time-step, the decision of which softmax layer to use choose adaptively made by an MLP which is conditioned on the context. We motivate our work from a psychological evidence that humans naturally have a tendency to point towards objects in the context or the environment when the name of an object is not known. We observe improvements on two tasks, neural machine translation on the Europarl English to French parallel corpora and text summarization on the Gigaword dataset using our proposed model.

1 Introduction

Words are the basic input/output units in most of the NLP systems, and thus the ability to cover a large number of words is a key to building a robust NLP system. However, considering that (i) the number of all words in a language including named entities is very large and that (ii) language itself is an evolving system (people create new words), this can be a challenging problem.

A common approach followed by the recent neural network based NLP systems is to use a

softmax output layer where each of the output dimension corresponds to a word in a predefined word-shortlist. Because computing high dimensional softmax is computationally expensive, in practice the shortlist is limited to have only top- K most frequent words in the training corpus. All other words are then replaced by a special word, called the *unknown word* (*UNK*).

The shortlist approach has two fundamental problems. The first problem, which is known as the *rare word* problem, is that some of the words in the shortlist occur less frequently in the training set and thus are difficult to learn a good representation, resulting in poor performance. Second, it is obvious that we can lose some important information by mapping different words to a single dummy token *UNK*. Even if we have a very large shortlist including all unique words in the training set, it does not necessarily improve the test performance, because there still exists a chance to see an unknown word at test time. This is known as the *unknown word* problem. In addition, increasing the shortlist size mostly leads to increasing rare words due to Zipf’s Law.

These two problems can be particularly critical in language understanding tasks such as factoid question answering (Bordes et al., 2015) where the words that we are interested in are often named entities which are usually unknown or rare words.

In a similar situation, where we have a limited information on how to call an object of interest, it seems that humans (and also some primates) have an efficient behavioral mechanism of drawing attention to the object: *pointing* (Matthews et al., 2012). Pointing makes it possible to deliver information and to associate context to a particular object without knowing how to call it. In particular, human infants use pointing as a fundamental communication tool (Tomasello et al., 2007).

In this paper, inspired by the pointing behav-